# LOGICAL REASONING
## For Junior and Senior Categories

## 1. SERIES & PATTERNS

A series is an ordered sequence governed by a hidden rule. Find the rule and then the missing/next term.

| Pattern type | How to identify | Typical move |
|---|---|---|
| Constant difference | Subtract consecutive terms | +5, −3 |
| Multiplication/division | Compare ratios | ×2, ÷3 |
| Increasing differences | Differences change regularly | +2,+4,+6 |
| Squares/cubes | Terms resemble $n^2$ or $n^3$ | 1,4,9,16 |
| Alternating | Odd/even positions differ | +2, ×2 |
| Repetition/cycle | A short block repeats | 2,5,2,5… |
| Letters | Use A=1…Z=26 | B,E,H = +3 |
| Mixed/visual | Separate components; track attributes | number + letter |

**Fast identification order**

- First differences → second differences → ×/÷ → odd/even positions → squares/cubes → letters/components → visual attributes.

**Worked examples**

**1.** 7, 12, 17, 22, ? Answer: 27 (+5)
**2.** 64, 57, 50, 43, ? Answer: 36 (−7)
**3.** 3, 6, 12, 24, ? Answer: 48 (×2)
**4.** 160, 80, 40, 20, ? Answer: 10 (÷2)
**5.** 4, 7, 11, 16, 22, ? Answer: 29 (+3,+4,+5,+6,+7)
**6.** 1, 4, 9, 16, 25, ? Answer: 36 (squares)
**7.** 2, 5, 4, 7, 6, 9, ? Answer: 8 (odd/even streams)
**8.** B, E, H, K, ? Answer: N (+3 letters)

**Series & Patterns — Practice**

| No. | Question | Answer |
|---|---|---|
| 1 | 9, 14, 19, 24, ? | 29 |
| 2 | 72, 64, 56, 48, ? | 40 |
| 3 | 5, 10, 20, 40, ? | 80 |
| 4 | 243, 81, 27, 9, ? | 3 |
| 5 | 2, 6, 12, 20, 30, ? | 42 |
| 6 | 3, 8, 15, 24, 35, ? | 48 |
| 7 | 1, 3, 6, 10, 15, ? | 21 |
| 8 | 50, 47, 41, 32, 20, ? | 5 |
| 9 | 2, 5, 4, 7, 6, 9, ? | 8 |
| 10 | 5, 8, 8, 11, 11, 14, ? | 14 |
| 11 | A, D, G, J, ? | M |
| 12 | Z, W, T, Q, ? | N |
| 13 | A2, C4, E6, G8, ? | I10 |
| 14 | 4, 8, 7, 14, 13, 26, ? | 25 |

| | | |
|---|---|---|
| 15 | 2, 3, 5, 8, 12, 17, ? | 23 |

## 2. CODING–DECODING

Coding changes a word, number, or symbol according to a fixed rule. Decoding reverses it.

| Variant | Key idea | Example |
|---|---|---|
| Letter shift | Move each letter by a fixed amount | CAT → DBU = +1 |
| Position coding | A=1, B=2,C=3…..Z=26 | DOG → 4-15-7 |
| Reverse alphabet | A↔Z, B↔Y… | H → S |
| Reversal | Write in opposite order | TRAIN → NIART |
| Rearrangement | Move blocks/positions | ABCDE → CDEAB |
| Alternating shift | Different shifts by position | +1,−1,+1… |
| Increasing shift | Shift grows by position | +1,+2,+3… |
| Substitution | Fixed symbol replacement | A=★, B=● |
| Conditional/mixed | Rule changes by category | vowels unchanged |

**Alphabet: A=1…Z=26. Reverse pairs: A–Z, B–Y, C–X, …, M–N.**

**Worked examples**

**1. CAT → DBU; DOG → ?** Answer: EPH (+1)

**2. DOG → CNF; CAT → ?** Answer: BZS (−1, A wraps to Z)

**3. BAT → DCV** Answer: +2 each letter

**4. CAT → ?** Answer: 3-1-20

**5. DOG position-sum** Answer: 26

**6. H under reverse alphabet** Answer: S

**7. TRAIN reversed** Answer: NIART

**8. BIRD with +1,−1,+1,−1** Answer: CJSC

**Coding–Decoding — Practice**

| No. | Question | Answer |
|---|---|---|
| 1 | If CAT→DBU, DOG→? | EPH |
| 2 | If CAT→BZS, DOG→? | CNF |
| 3 | If BAT→DCV, FOX→? | HQZ |
| 4 | Write LAMP in positions. | 12-1-13-16 |
| 5 | Position-sum of CAT? | 24 |
| 6 | Reverse-code P. | K |
| 7 | Reverse TRAIN. | NIART |
| 8 | If ABCDE→CDEAB, PQRST→? | RSTPQ |
| 9 | BIRD using +1,−1,+1,−1. | CJSC |
| 10 | LAMP using +1,+2,+3,+4. | MCPT |
| 11 | If A=★, B=●, C=▲, code ABC. | ★●▲ |
| 12 | Vowels unchanged; consonants +1: CAT. | DBT |
| 13 | Decode DBU under +1 coding. | CAT |
| 14 | Shift first letter by word length: CAT. | FAT |
| 15 | Adjacent position sums for CA, AT. | 4, 21 |

## 3. BLOOD RELATIONS

The safest method is to build a small family tree and answer from the requested point of view.

| Relation | Meaning |
|---|---|
| Father / Mother | Male/female parent |
| Son / Daughter | Male/female child |
| Brother / Sister | Male/female sibling |
| Grandfather / Grandmother | Parent of a parent |
| Uncle / Aunt | Sibling of a parent (or in-law, if stated) |
| Nephew / Niece | Sibling's son/daughter |
| Cousin | Child of an uncle or aunt |
| Maternal / Paternal | Mother's side / Father's side |

Worked examples

1. Ravi is son of Mina. Mina to Ravi? Answer: Mother
2. Mina is daughter of Kumar; Mina is mother of Ravi. Kumar to Ravi? Answer: Grandfather
3. Asha is sister of Bina; Bina is mother of Ravi. Asha to Ravi? Answer: Maternal aunt
4. John is father of Mary; Mary is sister of Tom. John to Tom? Answer: Father
5. Ali is brother of Sara; Sara is mother of Omar. Ali to Omar? Answer: Maternal uncle
6. A is mother of B; B is father of C. A to C? Answer: Grandmother
7. M is father of N; N is mother of O. M to O? Answer: Maternal grandfather
8. R is sister of S; S is father of T. R to T? Answer: Paternal aunt

**Blood Relations — Practice**

| No. | Question | Answer |
|---|---|---|
| 1 | P is mother of Q; Q is brother of R. P to R? | Mother |
| 2 | A is father of B; B is mother of C. A to C? | Maternal grandfather |
| 3 | X is brother of Y; Y is mother of Z. X to Z? | Maternal uncle |
| 4 | M is sister of N; N is father of O. M to O? | Paternal aunt |
| 5 | K is brother of L; L is brother of M. K to M? | Brother |
| 6 | D is daughter of E; E is son of F. D to F? | Granddaughter |
| 7 | G is wife of H; H is brother of J. G to J? | Sister-in-law |
| 8 | A is mother of B; B is sister of C. A to C? | Mother |
| 9 | S is son of T; T is daughter of U. S to U? | Grandson |
| 10 | R is father of S; S is father of T. R to T? | Grandfather |
| 11 | V is daughter of W; W is brother of X. V to X? | Niece |
| 12 | Y is sister of Z; Z is father of A. Y to A? | Paternal aunt |
| 13 | P is wife of Q; Q is brother of R. P to R? | Sister-in-law |
| 14 | L is mother of M; M is brother of N. L to N? | Mother |
| 15 | C is father of D; D is sister of E. C to E? | Father |

## 4. DIRECTION SENSE

Track movement and facing direction. Separate final position from the direction a person is facing.

| Rule | Meaning |
|---|---|
| N / S / E / W | Up / down / right / left on a map |
| Right turn | Clockwise 90° |
| Left turn | Anticlockwise 90° |
| Opposite | N↔S, E↔W |
| Diagonals | NE, NW, SE, SW |
| Displacement | Shortest straight distance from start to finish |
| Path length | Total distance actually travelled |

**Coordinate method: start at (0,0). East +x, West −x, North +y, South −y.**

**Worked examples**

1. 5 m north + 3 m east, Answer: NE
2. Facing north; turn right. Answer: East
3. Facing east; turn right. Answer: South
4. Facing south; turn left. Answer: East
5. Facing west; turn right. Answer: North
6. 6 m north + 6 m south. Answer: Back to start
7. 10 m west + 4 m east. Answer: 6 m west
8. 3 m north + 4 m east. Answer: 5 m displacement, NE

**Direction Sense — Practice**

| No. | Question | Answer |
|---|---|---|
| 1 | 7 N, then 4 E. Direction? | NE |
| 2 | Facing E, right twice. | W |
| 3 | Facing E, left. | N |
| 4 | Facing S, right. | W |
| 5 | 8 E and 6 N. Displacement? | 10 m NE |
| 6 | 10 W and 4 E. Final? | 6 m W |
| 7 | 6 N, 4 E, 6 S, 2 W. Final? | 2 m E |
| 8 | 7 S, 7 E, 7 N. Final? | 7 m E |
| 9 | 12 E, 5 N, 12 W. Final? | 5 m N |
| 10 | Facing N; right, right, left. | E |
| 11 | Facing W; left, left. | E |
| 12 | 5 N, 5 E, 5 S, 5 W. | Start point |
| 13 | 9 E, 12 N. Displacement? | 15 m |
| 14 | 4 S, 3 W. Displacement? | 5 m SW |
| 15 | Facing S; left, left, right. | E |

## 5. SYLLOGISMS

Use only the given statements. A conclusion is valid only when it must be true in every arrangement allowed by the premises.

| Statement | Venn interpretation |
|---|---|

| | |
|---|---|
| All A are B | A is inside B |
| No A is B | A and B do not overlap |
| Some A are B | At least one overlap exists |
| Some A are not B | At least one A lies outside B |

**Golden rules**

- All A are B ≠ All B are A.
- Separate "some" statements need not refer to the same person/object.
- Do not infer existence from "All" unless existence is given.
- Distinguish "must be true" from "may be true".
- A counterexample is enough to show that a conclusion is not necessary.

**Worked examples**

1. All A are B; all B are C. All A are C. Answer: Valid
2. All A are B. All B are A. Answer: Invalid
3. No A is B; some C are A. Some C are not B. Answer: Valid
4. All P are Q; some P are R. Some Q are R. Answer: Valid
5. All X are Y; no Y is Z. No X is Z. Answer: Valid
6. Some M are N; all N are P. Some M are P. Answer: Valid
7. All A are B; no C is B. No A is C. Answer: Valid
8. All R are S. Some R are S. Answer: Not necessarily

**Syllogisms — Practice**

| No. | Question | Answer |
|---|---|---|
| 1 | All A are B; all B are C. All A are C? | Valid |
| 2 | All A are B. All B are A? | Invalid |
| 3 | No A is B; some C are A. Some C are not B? | Valid |
| 4 | All P are Q; some P are R. Some Q are R? | Valid |
| 5 | All X are Y; no Y is Z. No X is Z? | Valid |
| 6 | Some M are N; all N are P. Some M are P? | Valid |
| 7 | All A are B; no C is B. No A is C? | Valid |
| 8 | All R are S. Some R are S? | Not necessarily |
| 9 | No A is B; some B are C. Some C are not A? | Valid |
| 10 | All D are E; some F are D. Some F are E? | Valid |
| 11 | Some G are H; some H are J. Some G are J? | Not necessarily |
| 12 | All L are M; all M are N; no N is O. No L is O? | Valid |
| 13 | Some P are Q; no Q is R. Some P are not R? | Valid |
| 14 | All T are U; no U is V. Some T are V? | Impossible |
| 15 | Some A are B; all B are C. Some A are C? | Valid |

# REASONING PRACTICE QUESTIONS

## I. SERIES & PATTERNS

**1.** 6, 13, 27, 55, 111, ?
**Ans:** 223
**2.** 4, 10, 22, 46, 94, ?
**Ans:** 190
**3.** 11, 15, 23, 35, 51, ?
**Ans:** 71
**4.** 2, 7, 17, 37, 77, ?
**Ans:** 157
**5.** 128, 121, 107, 86, 58, ?
**Ans:** 23
**6.** 3, 12, 60, 360, 2520, ?
**Ans:** 20160
**7.** 1, 5, 14, 30, 55, ?
**Ans:** 91
**8.** 96, 89, 75, 54, 26, ?
**Ans:** -9
**9.** 7, 14, 12, 24, 22, 44, ?
**Ans:** 42
**10.** 5, 9, 18, 34, 59, ?
**Ans:** 95
**11.** B, E, I, N, T, ?
**Ans:** A
**12.** Y, U, P, J, C, ?
**Ans:** U
**13.** C2, F6, J12, O20, U30, ?
**Ans:** B42
**14.** 2, 4, 12, 48, 240, ?
**Ans:** 1440
**15.** 9, 18, 16, 32, 30, 60, ?
**Ans:** 58
**16.** 1, 8, 27, 64, 125, ?
**Ans:** 216
**17.** 13, 19, 31, 49, 73, ?
**Ans:** 103
**18.** 4, 6, 12, 14, 28, 30, ?
**Ans:** 60
**19.** 10, 21, 43, 87, 175, ?
**Ans:** 351
**20.** D8, G12, K16, P20, V24, ?
**Ans:** C28

## II. CODING–DECODING

**21.** If LION → OLRQ, then RING → ?
**Ans:** ULQJ
**22.** If each letter is shifted +2, FROG → ?
**Ans:** HTQI

**23.** Using reverse alphabet coding (A↔Z, B↔Y...), CROWN → ?
**Ans:** XILDM
**24.** Reverse the word, then shift every letter +2. PLANT → ?
**Ans:** VPCNR
**25.** Odd-position letters +1, even-position letters −2. TIGER → ?
**Ans:** UGHCS
**26.** Shift letters by +1,+2,+3,+4,+5 respectively. CLOUD → ?
**Ans:** DNRYI
**27.** Vowels move A→E→I→O→U→A; consonants −1. MOUSE → ?
**Ans:** LUARI
**28.** Move the first two letters to the end. MONKEY → ?
**Ans:** NKEYMO
**29.** If the code of a word is the sum of the alphabet positions of its first and last letters, what is the code of BRIDGE?
**Ans:** 9
**30.** If A=4, B=8, C=12 ... Z=104, code BAD.
**Ans:** 8-4-16
**31.** Shift every letter −1, then reverse the word. TIGER → ?
**Ans:** QDFHS
**32.** Consonants +3; vowels −1. MARKET → ?
**Ans:** PZUNDW
**33.** Rearrange letters as 2nd, 4th, 1st, 3rd. TEAM → ?
**Ans:** EMTA
**34.** Double each alphabet position, reducing values above 26 cyclically. DOG → ?
**Ans:** HOC
**35.** Reverse-alphabet coding followed by +1. FISH → ?
**Ans:** VSI T
**36.** Shift letters successively +2,+3,+4,+5. CAGE → ?
**Ans:** EDKJ
**37.** Code = product of alphabet positions. What is the code of SUN?
**Ans:** 5586
**38.** Replace each letter by the letter 3 places ahead, then reverse. LAMP → ?
**Ans:** SPD O
**39.** Shift the 1st, 2nd, 3rd, 4th letters by +1,+2,+3,+4. BREAD → ?
**Ans:** CTHEE
**40.** Decode MNKU if the original word was first shifted +2 and then reversed.
**Ans:** SILK

## III. BLOOD RELATIONS

**41.** A is mother of B. B is brother of C. C is father of D. A is D's ?
**Ans:** Grandmother
**42.** P is father of Q. Q is sister of R. R is mother of S. P is S's ?
**Ans:** Grandfather
**43.** A is brother of B. B is mother of C. C is wife of D. D is father of E. A is E's ?
**Ans:** Maternal uncle
**44.** X is daughter of Y. Y is brother of Z. Z is father of W. X is W's ?
**Ans:** Niece

**45.** M is wife of N. N is sister of O. O is father of P. M is P's ?
**Ans:** Aunt
**46.** K is father of L. L is sister of M. M is mother of N. K is N's ?
**Ans:** Grandfather
**47.** P is daughter of Q. Q is brother of R. R is father of S. P is S's ?
**Ans:** Aunt
**48.** V is son of W. W is sister of X. X is mother of Y. V and Y are ?
**Ans:** Cousins
**49.** A is father of B. B is brother of C. C is mother of D. A is D's ?
**Ans:** Grandfather
**50.** R is mother of S. S is brother of T. T is father of U. R is U's ?
**Ans:** Grandmother
**51.** H is wife of J. J is brother of K. K is mother of L. H is L's ?
**Ans:** Aunt
**52.** M is father of N. N is sister of O. O is mother of P. M is P's ?
**Ans:** Grandfather
**53.** A is sister of B. B is father of C. C is brother of D. A is D's ?
**Ans:** Aunt
**54.** X is son of Y. Y is daughter of Z. Z is mother of W. X is W's ?
**Ans:** Grandson
**55.** P is mother of Q. Q is husband of R. R is mother of S. P is S's ?
**Ans:** Grandmother
**56.** F is brother of G. G is wife of H. H is father of J. F is J's ?
**Ans:** Uncle
**57.** T is father of U. U is sister of V. V is father of W. T is W's ?
**Ans:** Grandfather
**58.** A is mother of B. B is wife of C. C is father of D. A is D's ?
**Ans:** Grandmother
**59.** R is sister of S. S is father of T. T is mother of U. R is U's ?
**Ans:** Great-aunt
**60.** K is son of L. L is daughter of M. M is father of N. K and N are ?
**Ans:** Grandchildren / cousins by generation?
**Ans:** Grandson and son

## IV. DIRECTION SENSE

**61.** A moves 7 m N, 9 m E, 3 m S, 4 m W. Position from start?
**Ans:** NE, $\sqrt{41}$ m
**62.** Move 15 m W, 8 m S, 15 m E, 20 m N. Final position?
**Ans:** 12 m North
**63.** Facing North: right → right → left → right. Final direction?
**Ans:** South
**64.** 5 m N, 12 m E, 5 m S, 4 m W. Distance from start?
**Ans:** 8 m East
**65.** 20 m East and 21 m North. Shortest distance?
**Ans:** 29 m
**66.** Facing South-West, turn 90° clockwise. Direction?
**Ans:** North-West

**67.** 8 m N, 6 m W, 11 m S, 10 m E. Direction and distance?
**Ans:** SE, 5 m

**68.** 18 m S, 24 m E, 6 m N. Shortest distance?
**Ans:** $12\sqrt{5}$ m, SE

**69.** Facing East, turn right three times. Final direction?
**Ans:** North

**70.** 10 m N, 10 m E, 10 m S, 10 m W, 6 m N. Final position?
**Ans:** 6 m North

**71.** 14 m W, 7 m N, 5 m E, 7 m S. Final position?
**Ans:** 9 m West

**72.** Facing North-West, turn 180°.
**Ans:** South-East

**73.** 12 m N, 5 m W, 4 m S, 13 m E. Position?
**Ans:** NE, $8\sqrt{2}$ m

**74.** 9 m E, 40 m S, 9 m W, 10 m N. Position?
**Ans:** 30 m South

**75.** Facing South, turn left, right, right. Final direction?
**Ans:** West

**76.** 6 m W, 8 m N, 6 m E, 8 m S, 3 m N.
**Ans:** 3 m North

**77.** A is 8 m east of B. C is 6 m north of A. C is in which direction from B?
**Ans:** North-East

**78.** 11 m E, 7 m N, 3 m W, 7 m S.
**Ans:** 8 m East

**79.** Facing West, turn 135° clockwise.
**Ans:** North-East

**80.** 9 m S, 12 m W, 20 m N. Direction and distance?
**Ans:** North-West, $\sqrt{265}$ m

## V. SYLLOGISMS

For Q81–100, determine which conclusion(s) follow.

**81.** All artists are dreamers. All dreamers are thinkers.
I. All artists are thinkers.
**Ans:** I only

**82.** All engineers are graduates. Some graduates are athletes.
I. Some engineers are athletes.
**Ans:** Does not follow

**83.** No poets are careless. Some writers are poets.
I. Some writers are not careless.
**Ans:** Follows

**84.** All P are Q. No Q is R.
I. No P is R.
**Ans:** Follows

**85.** Some M are N. All N are O.
I. Some M are O.
**Ans:** Follows

**86.** All A are B. Some C are B.
I. Some C are A.
**Ans:** Does not follow

**87.** No X is Y. Some Z are X.
I. Some Z are not Y.
**Ans:** Follows

**88.** All doctors are trained. Some trained people are musicians.
I. Some doctors are musicians.
**Ans:** Does not follow

**89.** All cats are mammals. All mammals are animals. No animals are plants.
I. No cats are plants.
**Ans:** Follows

**90.** Some teachers are readers. No readers are lazy.
I. Some teachers are not lazy.
**Ans:** Follows

**91.** All P are Q. All R are Q. No Q is S.
I. No P is S.
II. No R is S.
**Ans:** Both follow

**92.** Some A are B. All B are C. No C is D.
I. Some A are C.
II. Some A are not D.
**Ans:** Both follow

**93.** All L are M. Some M are N. No N is O.
I. All L are M.
II. Some L are N.
**Ans:** I only

**94.** No drivers are careless. All racers are drivers. Some racers exist.
I. Some racers are not careless.
**Ans:** Follows

**95.** All X are Y. Some Y are Z. No Z is W.
I. Some Y are not W.
**Ans:** Follows

**96.** All A are B. Some B are C. No C is D.
I. Some A are C.
II. Some B are not D.
**Ans:** II only

**97.** Some P are Q. All Q are R. No R is S.
I. Some P are R.
II. Some P are not S.
**Ans:** Both follow

**98.** All musicians are artists. No artists are selfish. Some musicians exist.
I. Some musicians are not selfish.
**Ans:** Follows

**99.** All A are B. No B is C. Some D are C.
I. Some D are not B.
**Ans:** Follows

**100.** Some A are B. No B is C. All C are D.
I. Some A are not C. II. Some D are C. **Ans:** I only